Evidence-Governed Agent Harnesses

Making Evidence Survive Long-Running Agent Execution

What should an agent remember after a crash at step 40 of 60?
Not just its state—but what it verified, what it accomplished, and what it is still allowed to do.

The production problem

Execution state is not the same as evidence state

Plan — Tool Call — Result — Next Step

Crash at Step 40 of 60

What was verified?

What is still valid?

What is still authorized?

From state recovery to evidence-governed recovery

Execution State — Checkpoint — Crash — Resume

Evidence State — Verification History — Outcome / Authorization — Evidence-aware Resume

Recovery = execution continuity + evidentiary continuity

The core execution pattern

Evidence becomes part of the graph

Plan — Evidence Checkpoint — Tool Call — Result Verification — Evidence Update — Next Step

Not every step needs the same verification — consequential boundaries get stronger checks

Evidence as first-class runtime state

An evidence envelope carries more than a value

Identity / provenance	Validity / applicability
Observation time	Resource consumption
Evidence type	Outcome attribution
Verification status	Verification history

From “I have a value” → “I have evidence that is traceable, verified and applicable”

What survives a crash at step 40?

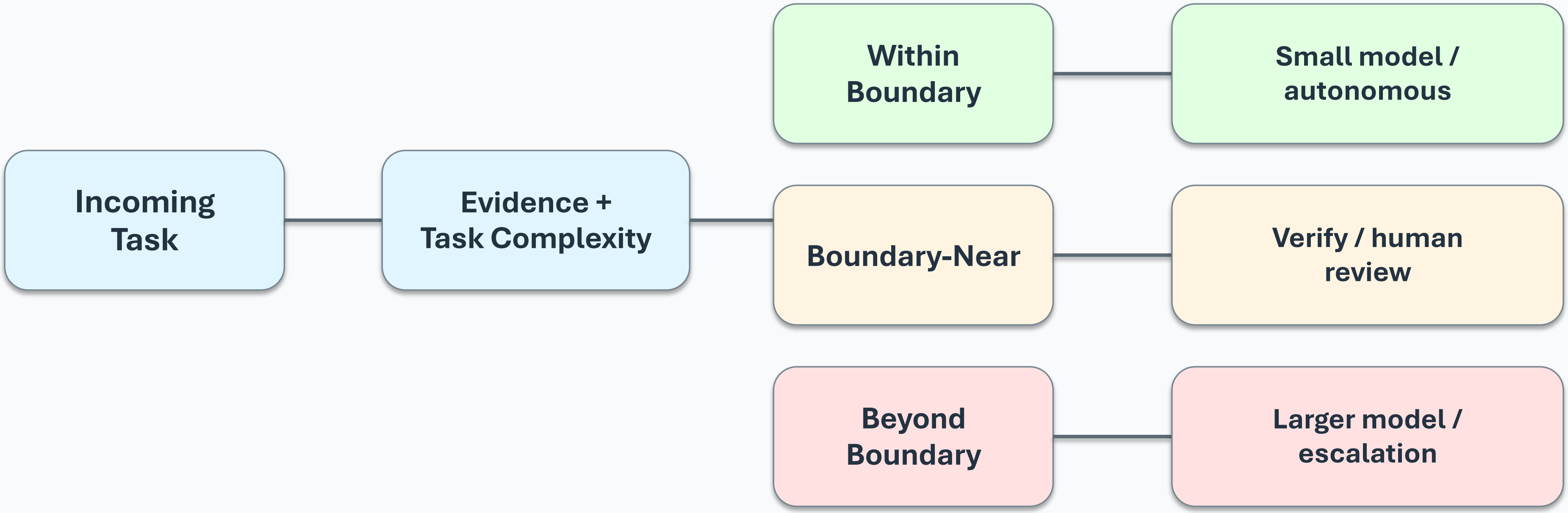
A production harness needs semantic continuity

Execution State	Where was the agent?
Evidence State	What did it know and verify?
Verification History	What checks already ran?
Outcome State	What was actually accomplished?
Authorization	What is the agent still permitted to do?

Resume only when the next action still has sufficient evidence and authorization

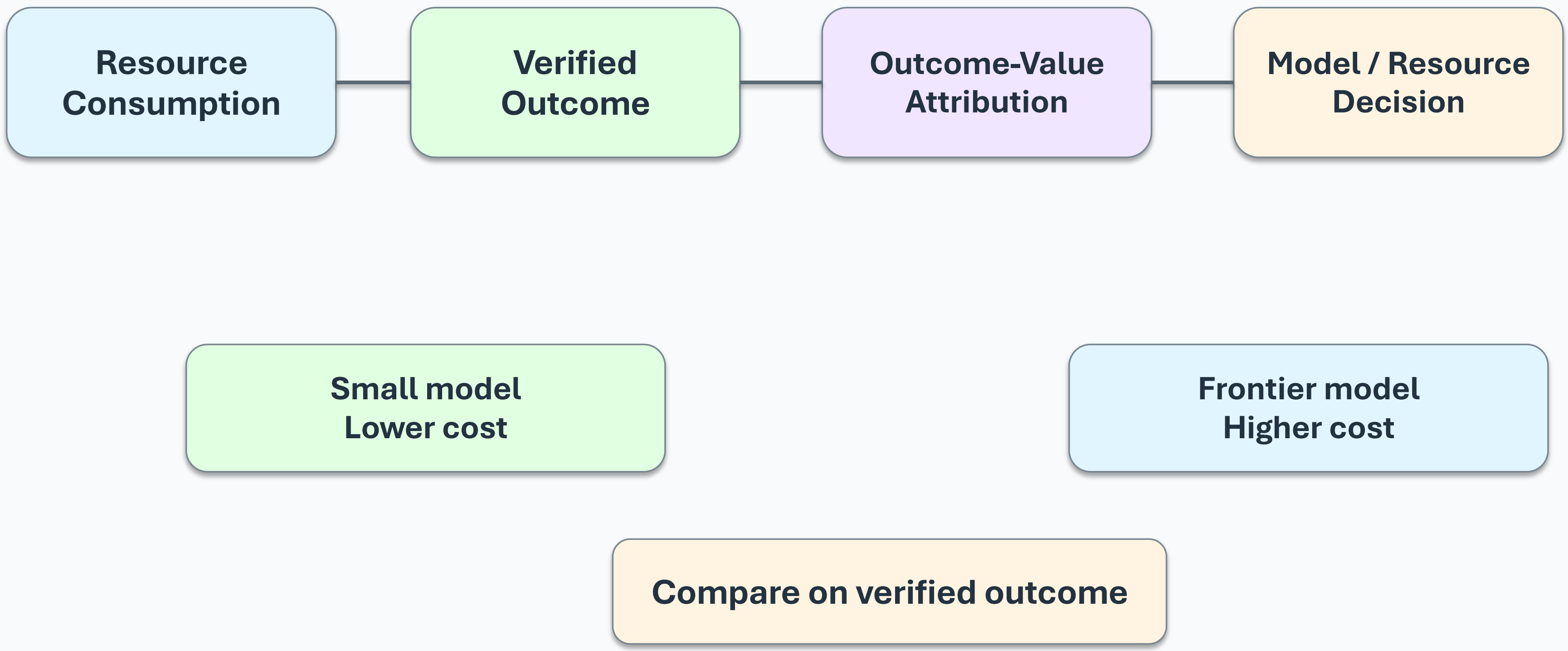
Small models: don't ask “Is 8B good enough?”

Ask where the model has an evidence-backed capability boundary



Resource and outcome accounting

Cost reduction is not success if the outcome degrades



Where verification belongs

Useful infrastructure vs ceremony

Not every step needs heavyweight verification	**Verification intensity should follow consequence and uncertainty**
Low-risk / reversible	**Continue**
Routine internal state	**Refresh evidence**
High-impact tool action	**Ask / human review**
Authorization-sensitive action	**Abstain / escalate**

What to evaluate before calling it production-ready

Measure the trade-offs

Checkpoint overhead	Latency / throughput impact
Recovery correctness	Evidence + execution restored consistently
Capability boundaries	Small-model task classification accuracy
Outcome / ROI accuracy	Consumption linked to verified outcomes

Goal: enough verification to improve reliability — without turning the harness into ceremony

What the engineering approach changes

Six practical principles

- State is not the same as evidence
- Evidence needs lifecycle semantics
- Verification belongs near consequential boundaries
- Small models need capability boundaries
- Recovery needs semantic continuity
- Governance should earn its place in the harness

What is demonstrated vs what remains to be validated

Research / design foundation	Implementation experiments	Production validation
Architecture	LangGraph integration	Real production workloads
Evidence-state model	8B boundary testing	Independent evaluation
Execution graph	Crash / resume experiments	Longitudinal evidence validity
Capability-boundary model	Overhead measurement	Operational failure modes
Resource/outcome accounting		

Do not turn a research design into a production claim — validate the failure modes first.

The broader production stack

Evidence sits alongside the harness, not above it

Agent / Model	Planning • generation • small / frontier models
Harness	Graph • loop • tools • permissions • retries
Evidence	Provenance • validity • verification • outcome
Observability	Telemetry • traces • cost • latency
Recovery	Checkpoint • durable state • replay • escalation

Key takeaways

What practitioners should leave with

- Treat evidence as first-class runtime state in long-running agents.
- Preserve evidence and verification history alongside execution checkpoints.
- Revalidate evidence at consequential boundaries instead of validating everything.
- Use evidence-backed capability boundaries to decide when smaller models are sufficient.
- Link model/resource consumption to verified outcomes, not cost alone.
- Measure the overhead and reliability benefit before adding another harness abstraction.

The harness should remember not only where the agent was — but why it was allowed to continue.

Thank You

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Fifth Elephant Hyderabad — Agent Harnesses & Small Models in Production